

Week 8 – Capstone Project: Real World Business Analysis

1. Project Overview

This project is a complete end-to-end business analysis project developed as the Week 8 Capstone Project for The Developers Arena Data Science Internship.

The objective is to analyze real-world business datasets, identify meaningful patterns, generate data-driven insights, and provide actionable business recommendations.

The project covers three major business areas:

- Sales Performance Analysis
- Customer Churn Analysis
- House Price Analysis

2. Objectives

The main objectives of this project are:

- Load and understand real-world datasets.
- Clean and prepare the data for analysis.
- Perform exploratory data analysis.
- Apply multiple data analysis techniques.
- Identify important business patterns and trends.
- Create professional data visualizations.
- Generate actionable business insights.
- Provide business recommendations.
- Prepare an implementation plan.
- Create a reproducible and portfolio-ready data analysis project.

3. Datasets

The project uses three datasets.

Sales Dataset

File:

`data/sales\_data.csv`

The dataset contains sales transaction information including:

- Date
- Product
- Quantity

- Price
- Customer_ID
- Region
- Total_Sales

The dataset is used to analyze product performance, regional performance and sales trends.

Customer Churn Dataset

File:

`data/customer\_churn.csv`

The dataset contains customer information including:

- CustomerID
- Tenure
- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod
- PaperlessBilling
- SeniorCitizen
- Churn

This dataset is used to understand customer churn and retention patterns.

House Prices Dataset

File:

`data/house\_prices.csv`

The dataset contains property information including:

- Property_ID
- Area
- Bedrooms
- Bathrooms
- Age
- Location
- Property_Type
- Price

This dataset is used to analyze property pricing patterns and factors affecting price.

4. Technologies Used

The project uses Python and the following libraries:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Scikit-learn
- Jupyter Notebook

5. Project Workflow

The project follows a complete data analysis workflow:

1. Business Problem Definition
2. Data Collection
3. Data Loading
4. Data Inspection
5. Data Cleaning
6. Exploratory Data Analysis
7. Statistical Analysis
8. Data Visualization
9. Business Insights
10. Recommendations
11. Implementation Plan
12. Final Documentation

6. Data Preparation

The datasets were inspected before analysis.

The following checks were performed:

- Dataset shape
- Column names
- Data types
- Missing values
- Duplicate records

- Numerical columns
- Categorical columns
- Date conversion where required

The data was cleaned and prepared before performing analysis.

Cleaned datasets are stored inside:

`cleaned\_data/`

7. Sales Analysis

Sales data was analyzed to understand overall business performance.

The analysis includes:

- Total sales
- Product-wise sales
- Region-wise sales
- Quantity analysis
- Sales trends over time
- Product performance comparison

Visualizations were created to make the results easier to understand.

The analysis helps identify high-performing products and regions and provides useful information for sales planning.

8. Customer Churn Analysis

Customer churn was analyzed to understand customer retention patterns.

The analysis includes:

- Overall churn rate
- Churn by contract type
- Churn by payment method
- Tenure analysis
- Monthly charges analysis
- Customer characteristics associated with churn

The purpose is to identify customer groups that may have a higher risk of leaving and to support customer retention strategies.

9. House Price Analysis

The house price dataset was analyzed to understand factors affecting property prices.

The analysis includes:

- Price distribution
- Area vs price

- Bedrooms vs price
- Bathrooms vs price
- Location-wise price comparison
- Property type analysis
- Correlation between numerical variables

The analysis helps understand important factors related to property pricing.

10. Analysis Techniques

Multiple analysis techniques were used throughout the project.

Exploratory Data Analysis

EDA was used to understand dataset structure, distributions, patterns and relationships.

Descriptive Statistics

Measures such as:

- Mean
- Median
- Minimum
- Maximum
- Standard deviation

were used to summarize the data.

Grouped Analysis

Group-by operations were used to compare:

- Product performance
- Regional sales
- Customer groups
- Contract types
- Property locations

Correlation Analysis

Correlation analysis was used to understand relationships between numerical variables.

Regression Analysis

Regression techniques were used where appropriate to understand relationships between relevant variables and support business interpretation.

11. Data Visualizations

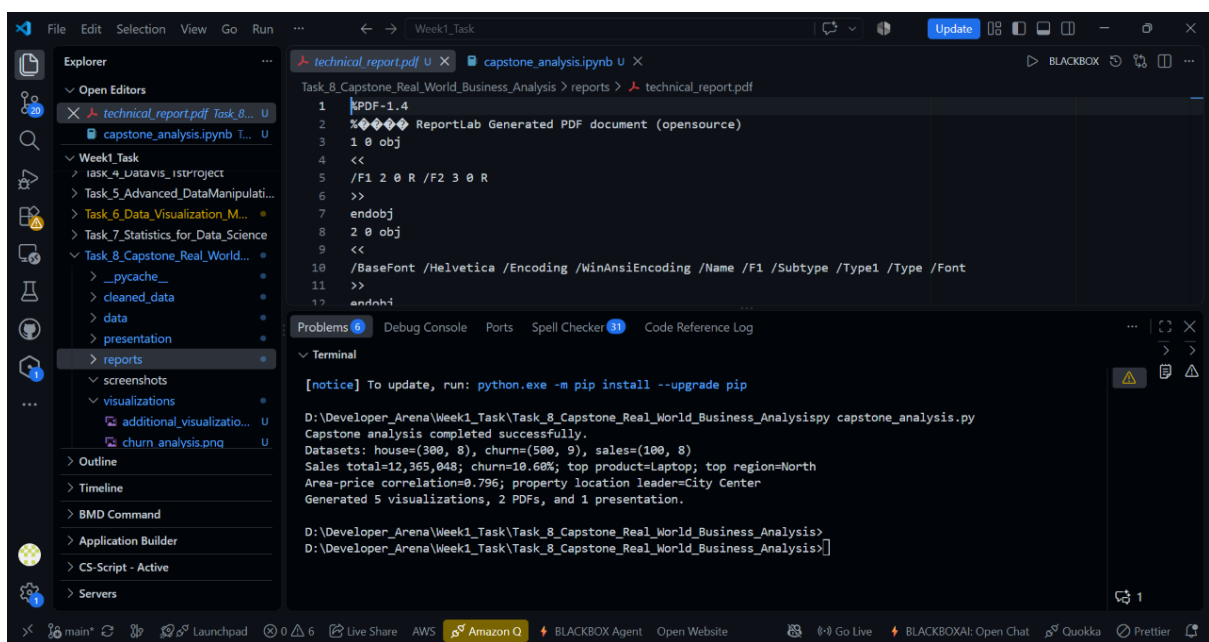
Professional visualizations were created and stored inside:

`visualizations/`

The visualizations include different chart types such as:

- Sales trend charts
- Product performance charts
- Regional analysis charts
- Customer churn charts
- Property price charts
- Correlation heatmaps

Each visualization is designed to communicate a meaningful business finding.



12. Business Insights

The analysis converts the technical results into business-oriented insights.

Key areas considered include:

Sales

Identify the products and regions contributing most to sales and identify sales trends.

Customer Churn

Identify customer groups with higher churn risk and understand factors associated with customer retention.

Property Prices

Identify important property characteristics and locations associated with differences in property prices.

The final insights are based on the calculated results from the datasets.

13. Business Recommendations

Based on the analysis, the following types of recommendations can be considered:

Sales Recommendations

- Focus on high-performing products.
- Identify opportunities in lower-performing regions.
- Use sales trends for inventory and sales planning.
- Monitor product performance regularly.

Customer Retention Recommendations

- Identify high-risk customer segments.
- Improve retention strategies for customers with higher churn probability.
- Review contract and payment patterns associated with churn.
- Develop targeted customer engagement strategies.

Property Business Recommendations

- Use location and property characteristics when developing pricing strategies.
- Identify property features that strongly influence price.
- Use historical pricing patterns to support better pricing decisions.

Recommendations should be implemented according to the actual findings generated by the analysis.

14. Implementation Plan

Priority	Action	Expected Impact
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High	Improve retention strategies for high-risk customers	Reduce customer churn
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High	Focus on high-performing products and regions	Improve sales performance
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Medium	Use data-driven property pricing	Improve pricing decisions
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Medium	Monitor business KPIs regularly	Better decision making
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Low	Update analysis periodically with new data	Continuous improvement
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15. Project Structure

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|— data/

| |— house_prices.csv

| |— customer_churn.csv

| |— sales_data.csv

|

|— cleaned_data/

- | |— cleaned_house_prices.csv
- | |— cleaned_customer_churn.csv
- | |— cleaned_sales_data.csv
- |
- |— visualizations/
- | |— sales_analysis.png
- | |— churn_analysis.png
- | |— property_price_analysis.png
- | |— correlation_heatmap.png
- | |— additional_visualization.png
- |
- |— reports/
- | |— executive_summary.pdf
- | |— technical_report.pdf
- | |— capstone_documentation.md
- |
- |— presentation/
- | |— business_analysis_presentation.pptx
- |
- |— screenshots/
- |
- |— capstone_analysis.ipynb
- |— capstone_analysis.py
- |— README.md
- |— requirements.txt